Supplementary Material for Recall-Constrained Influence-Aware User Retrieval for Product Recommendation

S1 Preliminaries and Framework Details

BPR-MF. BPR-MF learns user and item embeddings from implicit feedback by maximising the pairwise ranking objective

$$\max_{\Theta} \sum_{(u,i,j)\in D} \ln \sigma\big(\hat{x}_{ui} - \hat{x}_{uj}\big) - \lambda\|\Theta\|^2, \qquad \hat{x}_{ui} = \langle p_u, q_i\rangle, \tag{1}$$

where (u, i) is an observed interaction and j a sampled negative. After row normalisation the inner-product ranking coincides with the ranking induced by $-\|p_u - q_i\|^2$, so an L2 index is the correct retrieval structure for these embeddings.

HNSW. HNSW organises vectors into a hierarchy of proximity graphs. A query descends greedily through the upper layers and then runs a breadth-bounded search at level 0, controlled by ef. The returned list is approximate and its recall increases monotonically with ef, which is why a high-ef result serves as a proxy for exact retrieval in Section IV-B of the main text.

Influence maximization. Given G and a diffusion model, the task is to select K seeds maximising the expected number of reached nodes. The objective is monotone and submodular, so greedy selection attains a $(1 - 1/e)$ approximation, and CELF accelerates it by lazy evaluation. Our objective differs in two respects: coverage is counted over one hop, and the seed set must satisfy a retrieval constraint.

Offline and online stages. Offline, the framework reuses the existing HNSW index, stores the social graph in compressed sparse row form, and materialises a degree-ordered user list. Online, the candidate-pool module emits a pool that already contains influential users, the replacement module exchanges eligible anchors of the similarity Top-K, and the balance module bounds every exchange by the recall floor.

S2 Complete Notation Table

Table I lists every symbol used in the main text and in this supplement, grouped by the part of the framework it belongs to.

TABLE I
Complete notation, grouped by the part of the framework it belongs to.

Symbol	Description
Sets and graph	
U, I	user set and item set
V, E	node set and edge set of the social graph
$G = (V, E)$	undirected social graph
$N(v)$	one-hop neighbour set of user v
$\bar{d}$	average degree of G
Representation and retrieval	
p_u, q_i	embedding of user u and item i
d	embedding dimension
q	query item vector
$d(u, q)$	L2 distance between user u and query q
ef	HNSW search breadth
M, M_0	HNSW connectivity and its level-0 value
$NN_K(q)$	exact Top-K nearest neighbours of q
K	number of seed users to return
Objective and constraint	
S	returned seed user set, $\lvert S\rvert = K$
$f(S)$	one-hop coverage, $\lvert\bigcup_{v\in S} N(v)\rvert$
Recall@$K(S)$	$\lvert S \cap NN_K(q)\rvert/K$
τ	recall floor specified by the campaign
S_{proxy}	proxy truth set obtained by ANN search
min_overlap	required overlap with S_{proxy}, $\lceil K\tau\rceil$
slack	current_overlap − min_overlap
$I(\cdot)$	indicator function
Scoring and replacement	
α, A	dynamic balancing weight and its maximum
P	retrieval progress, $\lvert C\rvert/ef$
C	candidate set collected during retrieval
$inf(v, C)$	marginal coverage gain of v over C
ϵ	distance slack of the hard threshold λ
λ	hard distance threshold, $D_{\text{bound}}(1 + \epsilon)$
θ	soft threshold of the replacement rule
β	scale factor of θ

S3 Candidate-Pool Retrieval

Why static degree is insufficient. Degree ignores coverage overlap, so high-degree users that share followers are all scored highly; it does not change as the retrieval proceeds, so a node can keep attracting budget after its neighbours are covered; and it is unrelated to matching quality, so admitting it early pulls the search path out of the high-matching region. The marginal coverage gain, evaluated against the collected set C,

$$inf(v, C) = \Big|N(v) \setminus \bigcup_{u\in C} N(u)\Big|, \tag{2}$$

addresses all three.

Dynamic α. The first design switched α from 0 to A once the pool was full, which destabilises the search path. Making α a function of the progress $P = |C|/ef$ removes the jump. Four schedules were compared: step, linear $\alpha = A \cdot P$, quadratic

$\alpha = A \cdot P^2$, and a threshold schedule that stays at zero for the first half of the progress. Table II reports the comparison.

TABLE II
DIFFERENT $\alpha(P)$ SCHEDULES (SIFT1M, A=100, K=10, ef=100).

Schedule	Recall@10	Avg. Inf.	Lat. (ms)
Step	0.829	394.24	3.026
Linear	0.848	**403.09**	2.289
Quadratic	**0.849**	395.42	2.202
Threshold	0.845	395.66	2.194

All three continuous schedules outperform the step schedule, and the linear schedule gives the highest coverage while keeping recall at 0.848 and latency at 2.289 ms. It is therefore used throughout: early in the traversal α is near zero and distance dominates, which preserves the navigation path; late in the traversal α reaches A and coverage decides, which enlarges the influence potential of the pool.

Amplitude. Table III varies the amplitude of the linear schedule.

TABLE III
DIFFERENT A VALUES (SIFT1M, K=10, ef=100).

A	Recall@10	Avg. Inf.	Lat. (ms)
0	**0.987**	292.62	2.159
10	0.971	294.87	1.997
50	0.920	322.06	2.046
100	0.832	396.42	2.231
200	0.575	677.95	2.346
300	0.339	**945.21**	2.464

Recall decreases and coverage increases monotonically with A while latency stays between 2.0 and 2.5 ms, so the amplitude moves the operating point without changing the cost of the traversal. It is the quantity that the query-level adaptive mechanism adjusts at run time.

Workflow. The upper layers are traversed by pure physical greedy navigation. At level 0 the entry node initialises the visited array, a candidate min-heap keyed by hybrid score, a result max-heap, and a hash table of covered neighbours. Each iteration pops the lowest-score node, computes P and α, and evaluates the marginal gain of every unvisited neighbour; a neighbour is admitted only if the result heap is not full or its score beats the current worst. Admission increments the neighbour counters and eviction decrements them, so the coverage state always reflects the current set. The search stops when the candidate heap is empty, or when the best candidate is worse than the worst result while the result heap is full.

S4 HIERARCHICAL ITERATIVE REPLACEMENT

`1v1` rule. The initial set S is the HNSW Top-K and D_{bound} is its largest distance. A candidate is admissible only if $d_C \le \lambda$ with $\lambda = D_{\text{bound}}(1+\epsilon)$. For an admissible candidate and a seed $V \in S$,

$$\text{Gain}(C) = \Big| N(C) \setminus \bigcup_{u \in S} N(u) \Big|, \tag{3}$$

$$\text{Loss}(V) = \Big| \{ w \in N(V) : w \notin \bigcup_{u \in S \setminus \{V\}} N(u) \} \Big|, \tag{4}$$

and the exchange is accepted when the candidate is no farther and adds coverage, or when

$$\frac{\text{Gain}(C) - \text{Loss}(V)}{d_C - d_V} > \theta, \tag{5}$$

$$\theta = \beta \cdot \frac{\text{avg(Degree)}}{\text{avg(Dist)}}. \tag{6}$$

Oversampling. $3K$ candidates are fetched; the first K form the initial set and the remaining $2K$ enter the queue after the same filter λ, which gives the stage access to users outside the physical neighbourhood.

`2v2` fallback. When no single swap is accepted, the two seeds with the smallest marginal loss are removed and replaced by the candidate with the largest marginal gain together with the unselected candidate closest to the query. The acceptance test is unchanged.

TABLE IV
EFFECT OF β ON THE THREE LEVELS (SIFT1M, K=10, ϵ=0.2).

	`1v1`			+OS			+OS+`2v2`		
β	Rec.	Inf.	Lat.	Rec.	Inf.	Lat.	Rec.	Inf.	Lat.
0.2	0.067	1172.16	60.05	0.067	1178.43	59.86	0.074	**1206.55**	114.78
0.5	0.116	1105.05	43.80	0.112	1112.96	46.86	0.124	**1157.15**	87.40
0.8	0.149	1055.01	36.86	0.136	1074.11	40.68	0.154	**1104.04**	71.28
1	0.168	1014.85	33.48	0.148	1062.96	37.12	0.175	**1082.50**	63.51
2	0.293	848.11	22.90	0.259	890.26	25.18	0.305	**926.65**	42.96
5	0.544	617.16	13.56	0.519	638.40	14.08	0.537	**660.38**	25.99
10	0.768	399.72	7.14	0.748	406.78	7.97	0.736	**421.95**	14.51
20	0.909	324.65	4.70	0.888	348.58	5.51	0.883	**370.66**	10.22
50	0.963	306.72	4.50	0.956	312.00	4.69	0.956	**312.00**	8.25
100	0.980	294.86	4.28	0.976	296.01	4.52	0.976	**296.01**	8.05

Coverage increases strictly with the number of levels at every β, while recall stays within 0.02 of the `1v1` level, so the two optimisations raise the reachable coverage rather than shifting the recall–coverage curve. Latency grows with the number of accepted replacements. Small β makes the soft threshold permissive and recall falls to 0.07; β=100 leaves almost no room for replacement and the result approaches the plain ANN baseline. The main text uses β=20; in the complete framework the permitted amount of replacement is derived from the remaining slack instead.

S5 END-TO-END PROCEDURE

Algorithm 1 expands Algorithm 1 of the main text: it makes the state shared by the three modules explicit and states the termination condition of the replacement loop. The invariant that follows explains why the recall floor cannot be violated.

Invariant. Let $ov(S) = |S \cap S_{\text{proxy}}|$. Initially $ov(S) = K \ge m$. An exchange is executed only when VERIFY accepts it, and VERIFY rejects any exchange for which

$$ov' = ov - I(v^* \in S_{\text{proxy}}) + I(c^* \in S_{\text{proxy}}) \tag{7}$$

Algorithm 1 Recall-constrained influence-aware retrieval (expanded)

Require: q, K, τ, ef, A
Ensure: S with $|S| = K$ and $|S \cap S_{\text{proxy}}| \geq \lceil K\tau \rceil$
$S_{\text{proxy}} \leftarrow \text{ANN}(q, K, ef)$; $S \leftarrow S_{\text{proxy}}$
$m \leftarrow \lceil K\tau \rceil$; $cov \leftarrow \text{COUNTERS}(S)$
$C \leftarrow$ hybrid-guided retrieval with dynamic α (S3)
$Q \leftarrow \{c \in C : d(c, q) \leq \lambda\}$
repeat
    $v^* \leftarrow$ farthest eligible anchor of S
    $c^* \leftarrow \arg\max_{c \in Q} \text{Gain}(c)$
    if slack$(S) > 0$ **and** VERIFY(v^*, c^*) **then**
        $S \leftarrow S \setminus \{v^*\} \cup \{c^*\}$; update cov
    else if no admissible `1v1` exchange **then**
        try one `2v2` exchange; accept if coverage increases
    else
        break
    end if
until no further improvement **return** S

falls below m; the `2v2` fallback exchanges non-anchor members only and therefore leaves $ov(S)$ unchanged. Every reachable state satisfies the invariant, so the returned set meets the floor by construction. The invariant is stated against the proxy set, whose relation to exact neighbours is governed by the recall of the ANN search; this is why the proxy is calibrated offline against brute-force ground truth in the main text. The loop terminates because each accepted exchange increases coverage, coverage is bounded by $|V|$, and the number of accepted exchanges per query is bounded by $|C|$.

S6 Datasets, Preprocessing and Baselines

SIFT. 128-dimensional uint8 descriptors, four prefixes of 1M, 10M, 50M and 100M vectors sharing one social graph (average degree 10) and one set of 100 queries. Squared L2 distances over uint8 are integers below 2^{24}, so the integer and floating-point pipelines give bit-identical results. Ground truth is computed by brute force over the full corpus.

CiaoDVD. 16,043 users, 12,787 items and their trust relations; we keep ratings above 3.5, build a leave-one-out split, and learn 64-dimensional BPR-MF embeddings with 20 epochs, learning rate 0.05 and regularisation 10^{-4}. After projection onto users carrying at least one trust edge, the graph has 2,310 nodes and 21,400 undirected edges. Queries are items, candidates are users, and relevance is the exact K-nearest users of the item embedding.

KuaiRand-1K. 1,000 users, 7,388 candidate videos, 40,726 training impressions, 10,316 calibration impressions, 43,028 random-exposure test impressions, and a video co-click graph with 67,301 edges. For each impression a matching score is computed, candidates are sorted, and CTR@10, AUC and Logloss are measured against the real click labels.

Baselines. *HNSW* returns the Top-K of the graph index with ef=300 and receives no influence signal. *Degree-greedy* runs the same replacement with the same pool, oversampled queue, β and floor, and differs only in replacing the lowest-degree eligible anchor instead of the farthest one. *CELF-near* maximises one-hop coverage over the physical neighbours of the query; *CELF-global* does the same over the whole user set without a retrieval constraint.

Execution environment. A single node with 16 vCPUs and 60 GB of memory. The index is built serially (`OMP_NUM_THREADS=1`) so that the construction trajectory is deterministic, and queries are issued one at a time. The implementation is C++17, compiled with `g++ 11.2` at `-O3 -march=native`, with M=16 and $ef_{\text{construction}}$=200. No GPU is used.

S7 Complete Module Ablation

Table V reports the replacement module built up component by component, and Table VI compares each single module with the fused two-stage framework.

TABLE V
REPLACEMENT MODULE BUILT UP INCREMENTALLY (SIFT1M, K=10, β=20).

Configuration	Recall@10	Avg. Inf.	Lat. (ms)
Initial K + basic `1v1`	**0.909**	354.65	4.70
+ oversampled pool	0.888	368.58	5.51
+ `2v2` fallback	0.883	**370.66**	10.22

TABLE VI
SINGLE MODULES VERSUS THE FUSED FRAMEWORK (SIFT1M, K=10, A=200, β=20).

Configuration	Recall@10	Avg. Inf.	Lat. (ms)
Hybrid-score retrieval only	0.594	**688.02**	16.83
Replacement only	0.883	370.66	10.22
Full two-stage framework	0.818	436.25	**12.01**

Oversampling raises coverage by 3.9% for a 2.3% recall loss and 0.81 ms of latency, since the extra candidates are the influential users the physical Top-K lacks. The `2v2` fallback adds 0.56% with almost no recall loss, because it can leave states in which no single swap helps. In the fused framework, the retrieval-only variant reaches the highest coverage (688.02) but collapses recall to 0.594, while the replacement-only variant keeps recall at 0.883 but stalls at 370.66; the fused framework sits between them (0.818, 436.25) and has the lowest latency of the three, since the first stage already filters the pool the second stage examines.

S8 Influence-Maximisation Baselines

TABLE VII
COMPARISON WITH INFLUENCE-MAXIMISATION BASELINES.

	CELF-near		CELF-global		Ours	
Setting	Rec.	Inf.	Rec.	Inf.	Rec.	Inf.
SIFT 1M, K=10	0.490	240.50	0.000	1546.0	**0.700**	**612.29**
SIFT 1M, K=50	0.499	1162.78	0.000	7511.0	**0.700**	**3011.70**
SIFT 100M, K=10	0.488	252.27	0.000	1652.0	**0.699**	**662.28**
SIFT 100M, K=50	0.459	1190.53	0.000	8219.0	**0.698**	**3229.45**
CiaoDVD, K=10	0.630	47.88	0.000	720.0	**0.700**	**413.44**
CiaoDVD, K=20	0.684	85.48	0.002	981.0	**0.700**	**574.52**
CiaoDVD, K=50	0.727	190.00	0.004	1427.0	**0.701**	**877.90**

The two variants fail in opposite directions. CELF-global reaches 8219 coverage at K=50 on SIFT but returns a set that depends on the graph alone, so its recall is indistinguishable from zero. CELF-near keeps recall between 0.46 and 0.73 but its coverage is an order of magnitude below ours (240.50 against 612.29 at K=10), because the physical neighbours of an item are selected for similarity and rarely contain influential users. The proposed framework is the only configuration that satisfies a recall floor while improving coverage over the retrieval-only baseline.